WHAT IS PROGRAMMING?

Everything involved in making, controlling and communicating with digital devices. It involves writing a language that computers can understand.

PROGRAMMERS

A person who writes code for a device and is also involved in testing the code to ensure it is fully functional.

ADA LOVELACE

the first computer programmer. Worked with Charles Babbage to program the Analytical Engine.

LANGUAGES & JOBS

Roughly 700 programming languages in use today. Lots of jobs, including Software Engineers, Cyber Security Specialist. Also feeds into other career paths including Medicine and Marketing.

KEY SKILLS DEVELOPED

Analysis: Ability to extract information from a scenario in order to gain insight into the problem, allowing the creation rational solutions
Abstract thinking: ability to work with concepts rather than physical things
Pattern recognition: ability to see relationships and consequences from actions that haven't happened yet
Problem-solving: ability to find solutions to everything and anything, and be able to think outside the box in many situations

COMPILERS

The computer program that converts our 'human code' into the binary code computers can understand – we don't usually write in binary.

BENEFITS OF USING C#

Uses the .Net Framework, meaning it has benefits; Built in documentation | Very readable | IntelliSense (predictive text)| debugging tools readily available

INTEGRATED DEVELOPMENT ENVIRONMENTS

To write code, we use an IDE which has all the tools available to allow a programmer to view, write, test and run code.

PROGRAM STRUCTURE

```
1   using System;
2
3   namespace myFirstProgram
4   {
        0 references
5       class Program
6       {
            0 references
7           static void Main(string[] args)
8           {
9               Console.WriteLine("Hello World!");
10          }
11      }
12  }
```

This basically says "we're going to be using some premade tools"

This is a chapter within the book. Every chapter starts with this Main method.

This is one book in my series.

The { starts this specific book, and the } shows where that book ends.

Notice how it starts and ends within the series.

This section is like the name of my book series.

The { shows where the book series starts, and the } shows where the book series ends.

The first sentence.

DISPLAYING TEXT

Used when writing an output to the user

```
Console.WriteLine("Hello World!");
```

object | method | message | line terminator

INPUTTING TEXT

Used when asking the user to enter in data

```
string name;
name = Console.ReadLine();
```

The Program waits for user input

DATA TYPES

When using programming, it's important to tell a computer what type of data they are using in order to be able to handle it correctly. It also saves memory and prevents errors in complex calculations.

The most common data types are:

Data Type	What it holds
int	Whole numbers
string	text
double	Two decimal places
char	A single letter or symbol
bool	True or False

DECLARING VARIABLES

Variables are like storage boxes in RAM, and are used to hold data whilst a program is running. They follow the same pattern:

```
int score = 50;
```

data type — variable name — value

COMMENTS IN PROGRAMMING

Comments are annotations about code lines of code to know what the program is doing. It uses // to tell the program to ignore it

```
//displaying a welcome message
Console.WriteLine("Welcome");
```

SELECTION

Programming often uses SELECTION. This tells the computer to run a block of code **only if** a specific condition is true. It is called the IF statement.

```
string name = Console.ReadLine();

if(name == "Peter")
{
    Console.WriteLine("Hello Peter");
}
```

Within the brackets goes the condition

If the condition is true, then the code within the { and } runs.

The IF statement can also have an optional attachment called the ELSE statement. This runs if the condition equates to FALSE.

```
if (name == "Peter")
{
    Console.WriteLine("Hello Peter");
}
else
{
    Console.WriteLine("Hello Not Peter");
}
```

ITERATION

Loops allow us to write a block of code once and run it as many times as needed. This repetition is also called iteration.

Iterations can be programmed in three ways:

Count controlled | Condition controlled | Infinite

Count controlled loops, often using a FOR loop, repeat a block of code a known number of times.
Condition controlled loops, on the other hand, repeat until a condition is false. This condition is usually checked before the loop starts.

COUNT CONTROLLED LOOPS

Count controlled loops often use a FOR loop.
This repeats a block of code a set number of times. We know exactly how many times the block of code will run.

```
for(int i = 0; i < 10; i++)
```

Initialiser — Where the counter starts

Condition — How long to keep going

Iterator — How to update the counter

CONDITION CONTROLLED LOOPS

Often use a WHILE loop. Much like a count controlled loop, it has a condition to check before the program will run a section of code.

```
while(myName != "Mr B")
```

condition
Until this is false, the program will keep looping

INFINITE LOOPS

When writing WHILE loops, it is always a possibility of writing an infinite loop accidentally.
It's usually caused by the loop condition never being able to become false within the body of the loop.
There is a place where infinite loops are useful, however, such as a traffic lights systems which repeat forever.